MIL-R-26/1A
11 July 1967
SUPERSEDING
MIL-R-26/1
2 May 1966

MILITARY SPECIFICATION

RESISTORS, FIXED, WIRE-WOUND (POWER TYPE)

STYLES RW10, RW11, RW12, RW13, RW14, RW15, AND RW16

(NOT TO BE USED FOR NEW DESIGN)

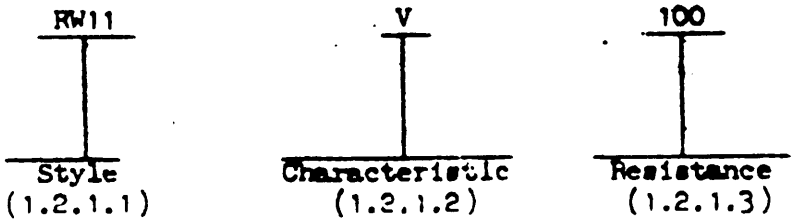
This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the detail requirements for styles RW10, RW11, RW12, RE13, RW14, RW15, and RW16 resistors. 1/

1.2 Classification.

1.2.1 Type designation. The type designation shall be in the following form:



1.2.1.1 Style. The style is identified by the two-letter symbol "RW" followed by a two-digit number; the letters identify power type, wire-wound, fixed resistors, and the number identifies the size, wattage rating, and construction of the resistors.

1.2.1.2 Characteristic. The characteristic is identified by a single letter that identifies the maximum continuous operating temperature (surface hot spot), the minimum insulation resistance value at the end of the moisture resistance test, and the resistance temperature characteristic in accordance with table I.

1/Not to be used for new design. For replacement purposes only.

FSC 5905

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TABLE I. Characteristic.

Symbol	Maximum continuous operating temperature 1/	Minimum insulation resistance at end of moisture resistance	Resistance temperat characteristic (percent per degree C)
	°C	Megohms	
V - - -	350	100	±.040 for resistance values below 20 ohms. ±.026 for resistance values of 20 ohms and above.

1/ This temperature is also the maximum permissible hot-spot surface temperature.

1.2.1.3 Resistance. The nominal resistance expressed in ohms is identified by a three-digit number; the first two digits represent significant figures and the last digit specifies the number of zeroes to follow. Where fractional values of an ohm are required, the letter "R" shall be substituted for one of the significant digits to represent the decimal point. When the letter "R" is used, succeeding digits of the group represent significant figures as shown in the following example:

R10 = 0.1 ohm
1R0 = 1.0 ohm

1.2.1.4 Center-tapped resistors. The letter "T" when added at the end of the type designation shall denote that the resistor is center-tapped.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-R-26 - Resistors, Fixed, Wire-wound (Power Type), General Specification for.

STANDARDS

MILITARY

MS90178 - Standard 24-Value Series Decade for Electronic Components of 5-, 10-, and 20-percent tolerances.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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3. REQUIREMENTS

3.1 Requirements. Requirements shall be in accordance with MIL R-26 only as specified herein.

3.2 Design and construction. Resistors shall be of the design, construction, and physical dimensions specified on figure 1, and as specified herein.

3.3 Power rating. The power rating shall be as specified in table II, based on full load operation at an ambient temperature of 25°C.

TABLE II. Power rating.

Resistor style	Power rating in watts
	Characteristic V
RW10 1/	203
RW11 1/	168
RW12 1/	125
RW13 1/	72
RW14 1/	58
RW15 1/	29
RW16	20

1/The power rating of center-tapped resistors shall be 90 percent of those shown.

3.4 Resistance. The minimum and maximum nominal resistance values shall be as specified in table III.

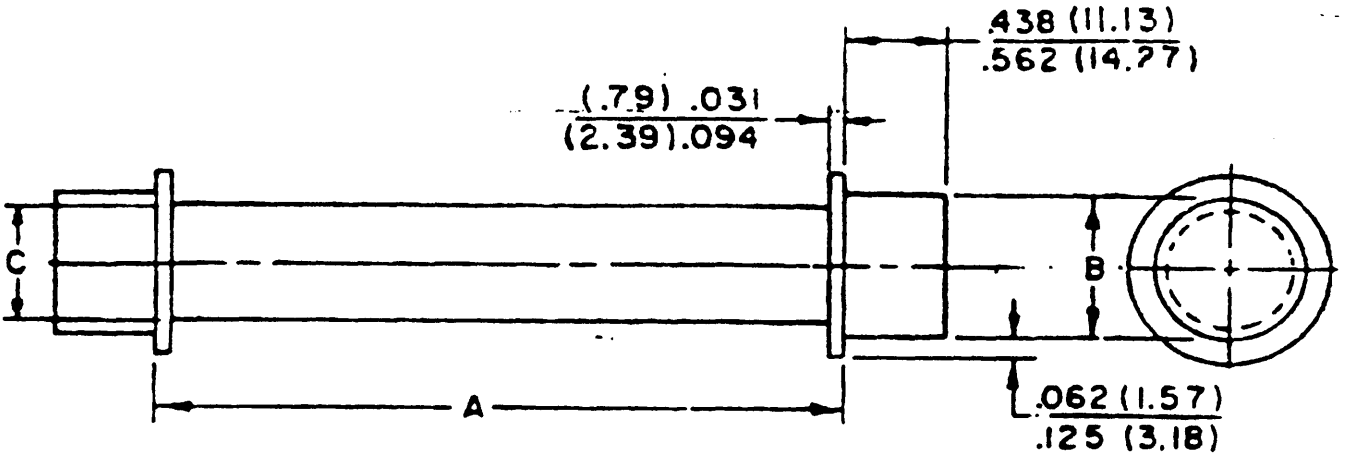
TABLE III. Minimum and maximum nominal resistance values.

Resistor style	Resistance value	
	Minimum	Maximum 1/
	ohm	
RW10	0.20	0.18 megohm
RW11	.10	.16 megohm
RW12	.10	.11 megohm
RW13	.10	51,000 ohms
RW14	.10	43,000 ohms
RW15	.10	15,000 ohms
RW16	.10	8,200 ohms

1/Based on use of 0.0017-inch nominal diameter wire.

3.4.1 Standard resistance values. For standard resistance values see MS 90178.

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		RW10	RW11	RW12	RW13	RW14	RW15	RW16
A	Min	10.375 (263.52)	8.562 (217.49)	6.375 (161.93)	4.062 (102.18)	3.375 (85.73)	1.875 (47.62)	1.312 (33.34)
	Max	10.500 (266.70)	8.688 (220.66)	6.500 (165.10)	4.188 (104.77)	3.500 (88.90)	2.000 (50.80)	1.438 (36.51)
B	Min	1.094 (27.79)	1.094 (27.79)	1.094 (27.79)	.781 (19.84)	.781 (19.84)	.531 (13.49)	.531 (13.49)
	Max	1.156 (29.36)	1.156 (29.36)	1.156 (29.36)	.844 (21.44)	.844 (21.44)	.594 (15.09)	.594 (15.09)
C	Min	1.000 (25.40)	1.000 (25.40)	1.000 (25.40)	.688 (17.48)	.688 (17.48)	.438 (11.13)	.438 (11.13)
	Max	1.312 (33.32)	1.312 (33.32)	1.312 (33.32)	1.062 (26.97)	1.062 (26.97)	.750 (19.05)	.750 (19.05)

Notes:
1. All dimensions in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information and are based upon 1 inch = 25.4 mm.
3. Millimeters are in parentheses.
4. The body diameter shall not be greater than the ferrule guard ring diameter.

FIGURE 1. Resistors, fixed, wirewound (power type) styles RW10, RW11, RW12, RW13, RW14, RW15, and RW16.

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3.5 Resistance tolerance. These resistors have a resistance range of 0.10 ohms to 0.10 megohms with the following resistance tolerances:

(a) Resistors of less than 1 ohm in resistance value shall have a tolerance of ±10 percent of the nominal resistance value. If center-tapped, the resistance tolerance between tap and terminal shall be ±10 percent.

(b) Resistors of 1 ohm and over in resistance value shall have a tolerance of ±5 percent of the nominal resistance value. If center-tapped, the resistance tolerance between tap and terminal shall be ±10 percent.

3.6 Center tapped resistors. Center tapped resistors are available in styles RW10, RW11, RW12, RW13, RW14, and RW15.

3.7 Thermal shock. The resistors shall not change in resistance in excess of ±(2 percent +0.05 ohm).

3.8 Short-time overload. The resistors shall not change in resistance in excess of ±(2 percent +0.05 ohm).

4. QUALITY ASSURANCE PROVISIONS

4.1 Qualification inspection. Not applicable to this specification.

4.2 General. The sampling, acceptance inspection, and test procedures specified in table II and table III of MIL-R-26 are applicable as sampling, inspection, and test procedures of this specification sheet.

4.3 Short-time overload. The maximum voltage shall be in accordance with table IV.

TABLE IV. Maximum voltage for short-time overload.

Resistor style	RW10	RW11	RW12	RW13	RW14	RW15	RW16
Voltage (volts, maximum)	10,100	8,200	6,100	3,800	3,100	1,600	1,000

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery shall be in accordance with MIL-R-26.

6. NOTES

6.1 Notes. In addition to the notes specified herein, the notes specified in MIL-R-26 are applicable to this specification.

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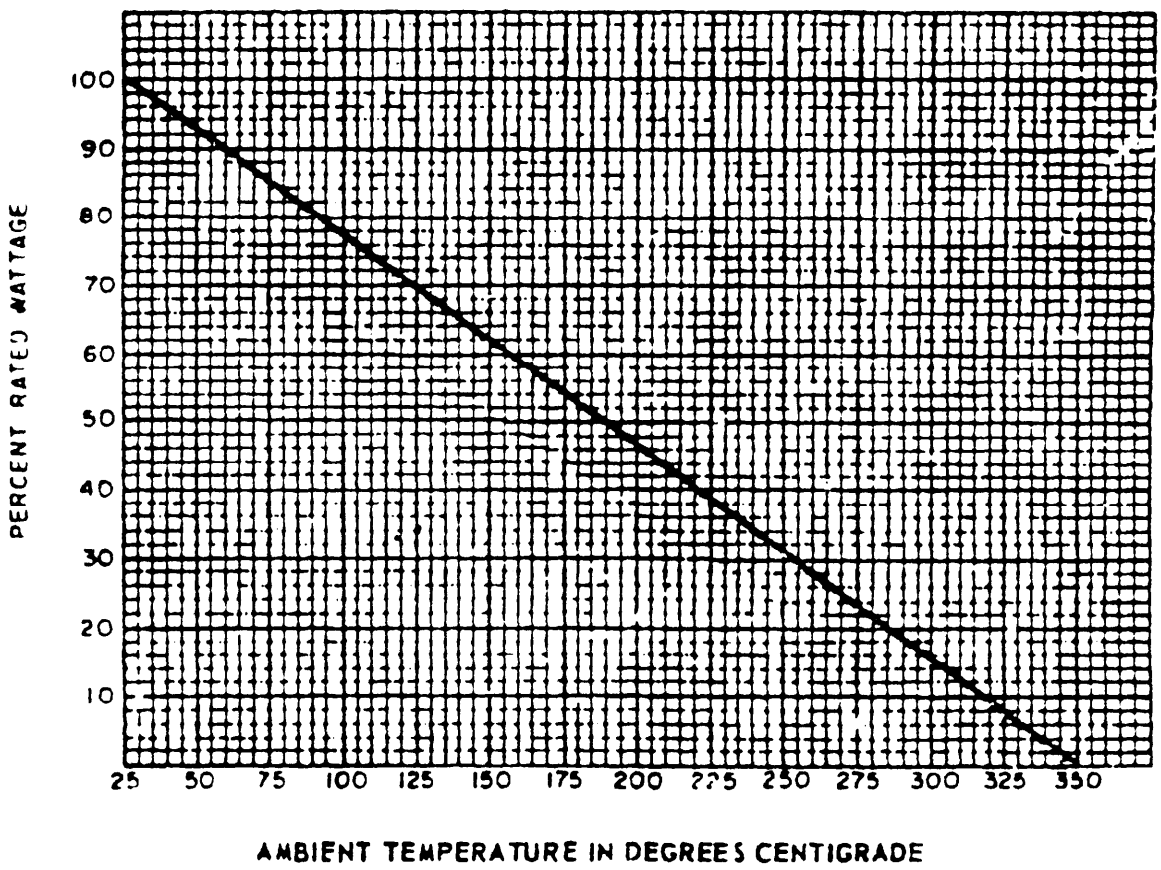


FIGURE 2. Derating curve for high ambient temperatures.

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6.2 Derating. Resistors shall be derated, when necessary, in accordance with figure 2.

6.3 Supersession data. This specification supersedes MIL-R-26/1, dated 2 May 1966, "Resistors, Fixed, Wire-wound (Power Type), Styles RW10 thru RW16."

6.4 Resistors covered by this specification are for replacement only and not for new design.

6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - EL
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Review activities:
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5. PROBLEM AREAS

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b. Recommended Wording:

c. Reason/Rationale for Recommendation:

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7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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